

بسته

$$\frac{1}{M} \sum_{j=1}^M \mathbb{P}_{\theta_j}(\rho(\hat{\theta}, \theta_j) \geq \delta) = \underbrace{Q(\rho(\hat{\theta}, \theta_J) \geq \delta)}_{\text{Take this for the next steps.}} = \sum_{j=1}^M \underbrace{Q(\rho(\hat{\theta}, \theta_j) \geq \delta | J=j)}_{Q(\rho(\hat{\theta}, \theta_j) \geq \delta | J=j)} \cdot \frac{1}{M}$$

Random Experiment: Pick a $J \in \{1, \dots, M\}$ uniformly at random.

Sample $\underline{Z} \sim \underline{\mathbb{P}}_{\underline{\theta}_J}(\cdot)$. Let Q be the distribution that $\underline{\mathbb{P}}_{\theta_j}(\rho \geq \delta)$

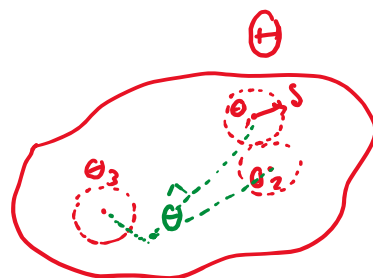
Characterize this experiment.

$$Q(Z, J) = \underbrace{Q(Z | J=j)}_{\mathbb{P}_{\theta_j}(Z)} \cdot \underbrace{Q(J=j)}_{\frac{1}{M}}$$

$$Q(Z) = \sum_{j=1}^M Q(Z | J=j) Q(J=j) = \frac{1}{M} \sum_{j=1}^M \mathbb{P}_{\theta_j}(Z)$$

$$\text{So: } m \geq \frac{1}{\delta} Q(\rho(\hat{\theta}, \theta^J) \geq \delta).$$

$$\text{Def. } \gamma(Z) := \underset{j \in [M]}{\operatorname{argmin}} \rho(\hat{\theta}, \theta_j)$$



$$\gamma(Z) = 3$$

$$\text{Claim: } \underbrace{\rho(\hat{\theta}, \theta_j)}_P < \delta \Rightarrow \underbrace{\gamma(Z)}_q = j.$$

$$\forall_{\substack{k \neq j \\ k \in [M]}} \underbrace{\rho(\hat{\theta}, \theta_k)}_P \geq \underbrace{\rho(\theta_j, \theta_k)}_{\geq 2\delta} - \underbrace{\rho(\theta_j, \hat{\theta})}_{< \delta} > \delta$$

$\Rightarrow \theta_j$ has the lowest dist. to $\hat{\theta}$ compared to other θ_k 's.

$$\Rightarrow \gamma(Z) = j.$$

$$\begin{aligned} P &\Rightarrow Q \\ \neg Q &\Rightarrow \neg P \end{aligned}$$

$$\begin{aligned} \mathbb{P}(P) &\leq \mathbb{P}(Q) \\ \mathbb{P}(\neg Q) &\leq \mathbb{P}(\neg P) \end{aligned}$$

$$Q(\underbrace{P(\hat{\theta}, \theta^T) \geq \delta}_{\neg p}) \geq Q(\underbrace{\Psi(z) \neq J}_{\neg q})$$

$$So \quad m \geq \Phi(\delta) \cdot Q(\underbrace{\Psi(z) \neq J}_{\neg q}).$$

$$\geq \inf_{\Psi} Q(\Psi(z) \neq J)$$

$$m \geq \Phi(\delta) \cdot \inf_{\Psi} \underbrace{Q(\Psi(z) \neq J)}_{\text{Prob. of error.}}$$

□

① Θ : 2δ -separated set. $N(\theta, \sigma^2)$ $\Theta = \mathbb{R}$ $\{0, 2\delta\}$
could be an option $\{0, 2\delta, -2\delta\}$

$$\textcircled{2} \quad \inf_{\Psi} Q(\Psi(z) \neq J).$$

Optimal
↳ Hypothesis Testing

Assume $\{\theta_1, \theta_2\}$ be the 2δ -sep. set.

$$\inf_{\Psi} Q(\Psi(z) \neq J) = \inf_{\Psi} \frac{1}{2} Q(\Psi(z) \neq 1 | J=1) + \frac{1}{2} Q(\Psi(z) \neq 2 | J=2)$$

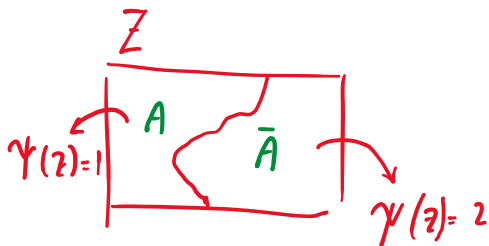
$$= \inf_{\Psi} \frac{1}{2} [P_{\theta_1}(\Psi(z) \neq 1) + P_{\theta_2}(\Psi(z) \neq 2)]$$

$$= \inf_A \frac{1}{2} [P_{\theta_1}(\bar{A}) + P_{\theta_2}(A)]$$

$$= \inf_A \frac{1}{2} [1 - P_{\theta_1}(A) + P_{\theta_2}(A)]$$

$$= \inf_A \frac{1}{2} \left[1 - \int_A P_{\theta_1}(z) dz + \int_A P_{\theta_2}(z) dz \right]$$

$$= \frac{1}{2} - \frac{1}{2} \sup_A \left\{ \int_A P_{\theta_1}(z) dz - \int_A P_{\theta_2}(z) dz \right\}$$



$$= \frac{1}{2} - \frac{1}{2} \sup_A \left| \int_A P_{\theta_1}(z) dz - \int_A P_{\theta_2}(z) dz \right|$$

$$=: TV(P_{\theta_1}, P_{\theta_2})$$

Side note:

$$\int_A P_{\theta_1} dz - \int_A P_{\theta_2} dz < 0$$

$$\int_{\bar{A}} P_{\theta_1} dz - \int_{\bar{A}} P_{\theta_2} dz > 0$$

$$= \cancel{\int P_{\theta_1} dz} - \cancel{\int P_{\theta_2} dz} + \int P_{\theta_2} dz$$

$$= \int P_{\theta_2} dz - \int P_{\theta_1} dz > 0$$

$$\Rightarrow n \geq \frac{1}{2} [1 - TV(P_{\theta_1}, P_{\theta_2})]$$

□

$$\text{Claim: } TV(P, q) = \frac{1}{2} \int |P(z) - q(z)| dz$$

$$TV(P, q) = \sup_A \left| \int_A \overset{q}{P}(z) dz - \int_A \overset{P}{q}(z) dz \right|$$

$$\text{Claim: } A^+ = \{z \mid P(z) > q(z)\}$$

$$A^- = \{z \mid q(z) > P(z)\}$$

$$\int_A \{P(z) - q(z)\} dz$$

< if $P < q \Rightarrow$

subopt. sol. as we already know that the sol. should be pos.

$$\Rightarrow TV(P, q) = \frac{1}{2} \left[\int_{P>q} (P-q) dz + \int_{q>P} (q-P) dz \right]$$

$$= \frac{1}{2} \int |P-q| dz$$

□

$$\text{Lemma: } \begin{matrix} P > 0 \\ q > 0 \end{matrix} \quad TV^2(P, q) \leq \frac{1}{4} \left[\int \frac{P^2}{q} dz - 1 \right].$$

$$\text{Proof: } TV^2(P, q) = \frac{1}{4} \left(\int |P-q| dz \right)^2 = \frac{1}{4} \left(\int \frac{|P-q|}{\sqrt{q}} \cdot \sqrt{q} dz \right)^2$$

$$\left(\int f \cdot g dz \right)^2 \leq \int f^2 dz \cdot \int g^2 dz \leq \frac{1}{4} \int \frac{(P-q)^2}{q} dz \cdot \underbrace{\int q dz}_1$$

$$= \frac{1}{4} \int \frac{P^2 + q^2 - 2Pq}{q} dz$$

$$= \frac{1}{4} \left[\int \frac{p^2}{q} dz + \underbrace{\int 4 dz}_1 - 2 \underbrace{\int p dz}_1 \right] = \frac{1}{4} \left[\int \frac{p^2}{q} dz - 1 \right] \quad \square$$

Example: $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$. $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$

$$\mathbb{E}\{\Phi(\hat{\theta}, \theta)\} = \mathbb{E}\{(\hat{\theta} - \theta)^2\}$$

$$= \text{Var}[\hat{\theta}] = \frac{1}{n^2} [\sum \sigma^2] = \frac{\sigma^2}{n}$$

Is there any other $\hat{\theta}$ s.t. $\mathbb{E}\{(\hat{\theta} - \theta)^2\} = o(\sigma^2/n)$?

$\{0, 2\delta\}$ be the 2δ -sep. set for $\Theta = \mathbb{R}$.

$$n \geq \Phi(\delta) \cdot [1 - \text{TV}(\mathbb{P}_0^n, \mathbb{P}_{2\delta}^n)]$$

$$\begin{aligned} \text{TV}^2(\mathbb{P}_0^n, \mathbb{P}_{2\delta}^n) &\leq \frac{1}{4} \left[\int \frac{(\mathbb{P}_{2\delta}^n)^2}{\mathbb{P}_0^n} dz - 1 \right] & \mathbb{P}_0^n &= \frac{1}{(\sqrt{2n}\sigma)^n} \exp\left\{-\frac{\|z\|^2}{2\sigma^2}\right\} \\ & & \mathbb{P}_{2\delta}^n &= \frac{1}{(\sqrt{2n}\sigma)^n} \exp\left\{-\frac{\|z - 2\delta\|^2}{2\sigma^2}\right\} \\ &= \frac{1}{4} \left[\int \frac{1}{(\sqrt{2n}\sigma)^n} \exp\left\{-\frac{\|z\|^2}{2\sigma^2} - \frac{4\|z\|^2}{\sigma^2} + \frac{4z^T c}{2\sigma^2}\right\} dz - 1 \right] \\ &= \frac{1}{4} \left[\int \frac{1}{(\sqrt{2n}\sigma)^n} \exp\left\{-\frac{\|z - 2c\|^2}{2\sigma^2} - \underbrace{\frac{4\|z\|^2}{\sigma^2} + \frac{4\|c\|^2}{\sigma^2}}_{\frac{4\|\delta\|^2}{\sigma^2}}\right\} dz - 1 \right] \end{aligned}$$

$$= \frac{1}{4} \left[e^{4n\delta^2/\sigma^2} - 1 \right]$$

$$n \geq \delta^2 \left[1 - \frac{1}{2} \sqrt{e^{4n\delta^2/\sigma^2} - 1} \right]$$

$$\delta = \frac{\sigma}{2\sqrt{n}}$$

$$n \geq \frac{\sigma^2}{4n} \cdot \underbrace{\left[1 - \frac{1}{2} \sqrt{e - 1} \right]}_{\approx 0.245 \geq 1/3} \geq \frac{\sigma^2}{12n} \Rightarrow \text{no } \hat{\theta} \text{ that has error } o(\frac{\sigma^2}{n}). \quad \square$$